MOHAMMAD ABU-SHELBAIA

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SUMMARY

Computer Engineering graduate from Birzeit University, working across embedded systems and software. Writes bare-metal C and ARM assembly for the TI TM4C123 (ARM Cortex-M4F) and teaches ENCS4110, the Computer Design Lab built on that platform, as a Teaching and Research Assistant. Python backend developer (Django, Flask, Docker, PostgreSQL) with industry experience at Silal; also builds automation tools and the department's lab-manual platform.

SKILLS

Languages:	C, Python, ARM Assembly, PIC Assembly, SQL, Bash
Embedded:	TI TM4C123 (Cortex-M4F), Bare-metal, Register-level programming, CMSIS, TivaWare, GPIO, Interrupts (NVIC), Timers, PWM, QEI, UART, ADC, LCD
Backend & DevOps:	Django, Flask, FastAPI, REST APIs, pytest, PostgreSQL, Docker, Nginx, Gunicorn, GitHub Actions
Tools:	Git, Linux, Make, arm-none-eabi-gcc, Keil uVision, OpenOCD

EXPERIENCE

Sep 2024 – Present
Teaching and Research Assistant — **Birzeit University**
- Teach 4+ sections per semester of ENCS4110, an ARM Cortex-M4 lab on the TI TM4C123 LaunchPad: GPIO, external and timer interrupts, LCD interfacing and ADC, in C and ARM assembly.
- Wrote the ENCS4110 lab manual used in these sections, published at encs4110.birzeit.me.
- Prepare practical exam scenarios, mentor students on assignments and projects, and grade their work.

Jul 2024 – Dec 2024
Backend Developer (Part-time) — **Silal**
- Developed and maintained a data pipeline for managing and analyzing product data.
- Wrote pytest unit tests for Flask REST API endpoints.

Mar 2024 – Jul 2024
Backend Developer Intern — **Silal**
- Built a full-stack bookstore application with Django and Tailwind CSS.
- Deployed it with Docker Compose behind Nginx and Gunicorn, with PostgreSQL.

PROJECTS

Web, VitePress
PWA, CI/CD
ECE Labs Platform — **birzeit.me**
Web platform that replaces the static PDF lab manuals of the Birzeit ECE department, now hosting three lab courses: searchable pages, copy-paste code, fixes that go live on merge through GitHub Actions, offline install, and printable PDFs built from LaTeX. Includes the ENCS4110 manual, written from scratch.

Embedded Systems
PIC16F877A, Assembly
PIC16F877A Calculator — **github.com**
PIC assembly calculator for the PIC16F877A with single-button input and a 16x2 LCD.

Concurrency
C, POSIX Threads
Ants Simulator — **github.com**
Multithreaded C simulation: a POSIX thread per ant, mutex-guarded state, SDL2 graphics.

Web Scraping
Python, iCalendar
Course Schedule — **github.com**
Built a Python CLI that logs into Ritaj, fetches the course schedule, and exports recurring iCalendar events merged with holidays and exam periods, detecting the current semester automatically.

EDUCATION

Sep 2020 – Jul 2024
B.Sc. Computer Engineering — **Birzeit University**
Graduated with Excellence (87%)
Coursework: Real-Time & Embedded Systems, Microprocessors, Computer Organization, Digital Design (Verilog), Computer Architecture

CERTIFICATIONS & LANGUAGES

Jul 2024
Data Analyst Nanodegree, Udacity — **Certificate**

Spoken
English – C2 (CEFR), Arabic – Native